

Program 7-Deploy a Web Application to Kubernetes using Minikube

Directory structure:

- Program7
- nginx-deployment.yaml

Step 1: Start Minikube

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/Program-7
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ ls
nginx-deployment.yaml
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ minikube start
🐳 minikube v1.37.0 on Linuxmint 22.2
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, dashboard, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$
```

minikube start
minikube status

Step 2: Create deployment file

nano nginx-deployment.yaml

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/Program-7
GNU nano 7.2 nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_/ Go To Line  M-E Redo
```

Step 3:Apply deployment

kubectl apply -f nginx-deployment.yaml

kubectl get deployments

kubectl get pods

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ nano nginx-deployment.yaml
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2            0           8s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
backup-job-29435360-mnmg6           0/1     Completed          0           23d
backup-job-29435365-f4rpk           0/1     Completed          0           23d
backup-job-29468485-mscd5           0/1     Completed          0           2m26s
my-cron-29435360-fgn6q              0/1     Completed          0           23d
my-cron-29435365-4znv2              0/1     Completed          0           23d
my-cron-29468485-vp5wg              0/1     Completed          0           2m26s
nginx-deployment-6f9664446b-n6d2r   1/1     Running            0           19s
nginx-deployment-6f9664446b-z2vs6   0/1     ContainerCreating  0           19s
nodejs-pod                          1/1     Running            6 (2m48s ago) 31d
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$
```

Step 4:Expose deployment

kubectl expose deployment nginx-deployment --type=NodePort --port=80

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/Program-7
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ nano nginx-deployment.yaml
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2            0           8s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl get pods
NAME                READY   STATUS              RESTARTS   AGE
backup-job-29435360-mnmg6    0/1     Completed          0          23d
backup-job-29435365-f4rpk    0/1     Completed          0          23d
backup-job-29468485-mscd5    0/1     Completed          0          2m26s
my-cron-29435360-fgn6q      0/1     Completed          0          23d
my-cron-29435365-4znv2      0/1     Completed          0          23d
my-cron-29468485-vp5wg      0/1     Completed          0          2m26s
nginx-deployment-6f9664446b-n6d2r    1/1     Running            0          19s
nginx-deployment-6f9664446b-z2vs6    0/1     ContainerCreating  0          19s
nodejs-pod              1/1     Running            6 (2m48s ago)    31d
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl expose deployment nginx-deployment
--type=NodePort --port=80
service/nginx-deployment exposed
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP    23d
nginx-deployment    NodePort    10.104.20.56 <none>        80:31678/TCP 12s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31678

🔗 Opening service default/nginx-deployment in default browser...

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-7$
```

Step 5:Access app

kubectl get services

minikube service nginx-deployment

